REWARD ERRORS BECOME PERSISTENT STATE: A CONTROLLED INTERVENTION IN REWARD-CONDITIONED AGENT MEMORY

Anonymous authors
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ABSTRACT

Many external-memory agents use terminal feedback not only to score an episode but also to choose how its trajectory is summarized into persistent memory. This creates a state-update channel for evaluator error. We isolate that channel in released ReasoningBank/WebArena Shopping trajectories by holding each source trajectory byte-identical and counterfactually flipping only the success/failure reflection branch. Among four pairs complete in both conditions, every memory changes (mean token-set Jaccard distance 0.735); two failure-branch writes are provider-incomplete, so this result is conditional on paired completion. On eight disjoint trajectories, reward-mode divergence exceeds a stronger same-mode semantic-preserving rewording control (0.700 vs. 0.595; paired excess 0.105, exact $p = 0.0078$). The paired states alter 7/12 aligned next actions. A same-support 4×4 terminal replication after an initial non-pass, with the same 0.15/$p < 0.05$ dual gate, yields mean absolute success-rate difference 0.15625 ($p = 0.00074$) but strong source–future heterogeneity. A no-memory control shows that neither reward-conditioned branch is uniformly beneficial or harmful. A GLM-5.3 writer reproduces write-time divergence but its terminal replication misses the practical-effect floor (0.140625), with two sign reversals. Thus the robust result is the write-time causal channel; downstream magnitude and direction are configuration-dependent.

1 INTRODUCTION

External-memory agents can turn the end of one episode into persistent state for the next. A completed trajectory is summarized, stored, and later retrieved as context, often without any parameter update. When terminal feedback selects how that summary is written, the evaluator is therefore part of the state-update mechanism rather than only a scorer. An erroneous label can cause the same experience to be committed through a different reflection objective and remain available to future decisions.

Several neighboring literatures establish the pieces of this pipeline without identifying this write-time channel. JudgeBench and RewardBench document failures of LLM judges and reward models (Tan et al., 2025; Lambert et al., 2024). ReasoningBank and Reflexion already turn success/failure feedback into reusable textual memory (Ouyang et al., 2026; Shinn et al., 2023); Mem-α and AttriMem go further by learning memory-construction policies from outcome or process reward (Wang et al., 2025; Li et al., 2026). Memory Reward Inflation and RoMeRL study how reward affects valuation and reuse of memories that already exist (Asadolahi et al., 2026; Yang et al., 2026), while recent fragility work measures run variance and task-order sensitivity (Ye et al., 2026). We do not claim judge unreliability, feedback-driven memory construction, memory-reward amplification, or run variance as new. The missing object is earlier and more controlled: when the construction policy and experience are fixed, can a terminal-label error select a different durable memory state, and can that state difference propagate under matched future evidence?

We study this as a paired causal intervention. For each source episode, the task and full trajectory are held byte-for-byte identical while the terminal label counterfactually selects the released success- versus failure-reflection branch. Because those branches necessarily use different instructions, simple text divergence is not enough: generic prompt sensitivity is the strongest reduction. We therefore compare the reward-mode intervention against semantic-preserving rewordings within the same reflection mode whose surface perturbation is deliberately larger. At reuse time, future task evidence and policy configuration are fixed while only the persistent memory is swapped, and every complete source-memory pair is crossed with every frozen future task rather than selecting favorable cells.

The evidence supports one causal chain. First, all four source pairs complete in both writer conditions produce different memory content and operation-slot structure (mean token-set Jaccard distance 0.735); two additional failure-branch writes terminate at the provider layer before assistant text exists, so the write estimand is explicitly conditional on paired completion. Second, on eight disjoint trajectories, reward-mode divergence exceeds the stronger same-mode wording control (0.700 versus 0.595; paired excess 0.105, exact $p = 0.0078$). Third, the resulting states reach later behavior: 7/12 aligned fixed-state probes change the next structured action, and a same-support 4×4 terminal replication with eight fresh rollouts per condition yields mean absolute success-rate difference 0.15625 with permutation $p = 0.00074$. The terminal effect is heterogeneous rather than uniformly harmful: 8/16 cells are zero and 84.1% of centered signed-effect variation lies in the source–future interaction residual.

The boundaries sharpen the mechanism rather than broaden the claim. A fresh no-memory control shows that the success-conditioned branch is not uniformly better and the failure-conditioned branch is not uniformly worse. A second GLM-5.3 writer again changes all four paired memories, but its matched terminal replication reaches 0.140625 and misses the frozen 0.15 practical-effect floor despite $p = 0.00012$, with two direction reversals. Thus write-time state divergence transfers more robustly than downstream magnitude or direction. Finally, the released ReasoningBank retriever is top-1 over label-invariant task-description embeddings, so a proposed multi-bit full-bank corruption interaction collapses to the single retrieved entry; we stop that sweep by matched simplification rather than change the substrate to manufacture an interaction.

Contribution and closest-work boundary. The contribution is the identification of a single reward-error $\rightarrow$ persistent-state $\rightarrow$ future-behavior channel under a fixed memory-construction policy. Its load-bearing pieces are the byte-identical write intervention, the stronger reward-preserving wording reduction, and the matched downstream swap over the full frozen source–future support. The no-memory and cross-writer results locate where this chain does and does not generalize; the corruption-rate calculation is only a bounded consequence analysis of measured conditional effects. We therefore make no claim of a new memory architecture, a new reward-robustness method, a universal disadvantage of failure-conditioned memory, writer-invariant downstream effects, or a general law of reward-corruption variance.

2 Related Work

Evaluator reliability. JudgeBench and RewardBench establish that LLM judges and reward models can be unreliable before their outputs are used for adaptation (Tan et al., 2025; Lambert et al., 2024); Plan-RewardBench exposes additional brittleness on tool-integrated trajectories (Wang et al., 2026). These works motivate the possibility of a wrong terminal label, but their object ends at scoring accuracy. Our object begins when that label is consumed by a persistent-state update.

Feedback-conditioned memory construction. ReasoningBank explicitly distills different memories from successful and failed trajectories (Ouyang et al., 2026); Reflexion stores evaluative verbal feedback for later trials (Shinn et al., 2023); Agent Workflow Memory extracts reusable procedures from interaction histories (Wang et al., 2024). Mem-α learns what to store and update from downstream reward, and AttriMem adds attribution-derived process feedback to learn a memory-construction policy (Wang et al., 2025; Li et al., 2026). We therefore surrender any novelty claim that feedback can shape memory or that success/failure-conditioned writers are new. Our residual treatment fixes the existing construction policy and the source trajectory, then counterfactually changes only which released reward-conditioned branch writes the durable state.

Memory valuation, reuse, and self-improvement variance. Memory Reward Inflation studies how imperfect proxy reward can overvalue erroneous memories and amplify them through reuse (Asadolahi et al., 2026). RoMeRL addresses misleading trajectory-level utility updates and the memory-reward trap (Yang et al., 2026). Ye et al. quantify run variance, curriculum sensitivity, and memory

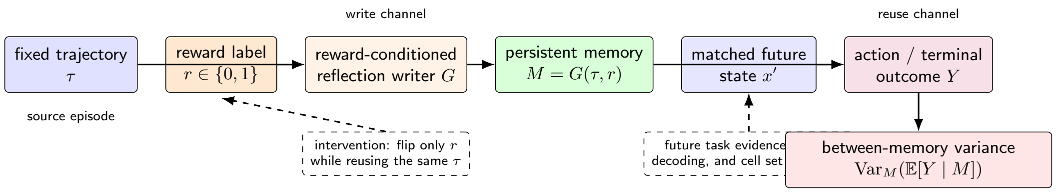

Figure 1: The reward-to-memory-to-outcome channel. The source trajectory is fixed while the reward label selects the reflection writer. The resulting persistent memory is then injected into a matched held-out task packet under a fixed policy configuration. Repeated outcomes identify the between-memory component of run variance.

underspecification in self-improving agents (Ye et al., 2026). These are the closest downstream neighbors, but their intervention is valuation, utility update, retrieval/reuse, or system-level variation. We instead isolate the earlier write boundary and then carry the resulting state through a matched future-evidence swap. Accordingly, our plug-in variance curve is not presented as a new variance phenomenon or empirical corruption sweep.

Defended residual. The narrow novelty is an identification object rather than a new memory algorithm: a byte-identical experience, a reward-conditioned writer-mode intervention under a fixed construction policy, a stronger same-mode wording reduction, and a fully crossed downstream memory swap under fixed future evidence. This distinguishes the paper from both "reward can be wrong" and "feedback can change memory" while keeping the claim local to the released ReasoningBank/WebArena configuration.

3 Causal Object: Reward Error as a Persistent-State Intervention

Let task x produce trajectory τ, and let the terminal evaluator emit $r \in \{0,1\}$. A reward-conditioned memory writer constructs

$$M(r) = G(\tau, r). \tag{1}$$

In the released ReasoningBank implementation, $r = 1$ selects the success-reflection objective and $r = 0$ selects the failure-reflection objective. The intervention studied here holds τ fixed and changes only this branch. The immediate causal estimand is therefore the paired persistent-state contrast $M(1)$ versus $M(0)$ for the same experience.

For a matched future task state x', let Y denote a downstream behavioral or terminal outcome under retrieved memory M. The reuse-stage contrast is

$$\Delta_Y(x') = \mathbb{E}[Y \mid do(M = M(0)), x'] - \mathbb{E}[Y \mid do(M = M(1)), x']. \tag{2}$$

The future task evidence and policy configuration are held fixed across the pair. This separates variation introduced by persistent memory state from variation in the observed future state.

3.1 A bounded variance consequence

If terminal-label corruption stochastically selects between the two measured reward-conditioned memories, the resulting state mixture has a direct variance consequence. Let M denote the persistent memory state available to later tasks. The law of total variance gives

$$\mathrm{Var}(Y) = \mathbb{E}_M[\mathrm{Var}(Y \mid M)] + \mathrm{Var}_M(\mathbb{E}[Y \mid M]). \tag{3}$$

The second term is the component induced by variation in persistent memory state. If reward corruption changes which memory is written and the paired memories produce different conditional success probabilities, reward uncertainty contributes directly to this term. Section 7 estimates the paired conditional effect and converts it into an explicit corruption-rate variance curve.

This consequence is secondary to the causal identification above. No parameter update is required for the channel itself: in the released external-memory mechanism, a mislabeled episode can change the context available to a later retrieval.

4 EXPERIMENTAL SETUP

Frozen units and completion-conditioned estimand. We use released WebArena Shopping task metadata and released ReasoningBank trajectories rather than collecting a new self-improvement stream. The write intervention requests six source trajectories balanced across original benchmark outcomes. Four produce complete memory outputs under both reward-conditioned writer branches; two failure-branch calls terminate at the provider response-state layer before assistant text exists. We neither re-POST nor impute those missing arms. Consequently, the paired write estimand and every downstream source-memory analysis are conditional on the four complete pairs, and condition-dependent provider completion cannot be ruled out. The prompt-sensitivity control uses eight disjoint trajectories. Terminal evaluation crosses all four complete source-memory pairs with all four frozen future tasks, yielding 16 cells with no outcome-driven source/task selection.

Models, controls, and estimands. DeepSeek-V4-Flash is the primary memory writer at temperature zero; Doubao Seed 2.0 Mini is the downstream terminal policy at temperature 0.2. Write-stage observables include token-set and deterministic operation-slot divergence. The strongest write-stage reduction is a semantic-preserving same-mode rewording whose lexical perturbation exceeds the released success/failure prompt difference. Reuse-stage evidence holds future prompt, browser evidence, policy, and evaluator fixed while swapping only memory. The primary terminal statistic is the mean absolute success-rate difference over the full 16-cell support, with the frozen dual gate $|\Delta| \geq 0.15$ and within-cell permutation $p < 0.05$. A source-independent no-memory arm is used only to locate which reward-conditioned branch departs from omission. A sequential GLM-5.3 replication tests whether the write channel and downstream magnitude transfer across writer families under the unchanged terminal gate.

Execution reliability. All scientific stages are inference-only: there is no training or local GPU fine-tuning. Provider outputs are content-addressed before analysis, execution-invalid attempts have zero scientific authority, and incomplete scientific units are never imputed. Appendix D reports stage-level request accounting, including the failed GLM parent attempt whose exact POST total is not reconstructible after an execution-layer concurrency race.

5 WRITE-TIME INTERVENTION: DOES THE REWARD LABEL CHANGE PERSISTENT MEMORY?

We begin with a paired intervention over released WebArena Shopping trajectories from the Salesforce fragility study. A deterministic sampler requests six source trajectories, balanced across original benchmark outcomes. For each trajectory, the task text and full action history are copied byte-for-byte into two ReasoningBank writer calls. One call uses the released success-reflection instruction and the paired call uses the released failure-reflection instruction. DeepSeek-V4-Flash writes memory at temperature zero, and raw provider responses are content-addressed before analysis.

Four of the six requested trajectory pairs return complete memory outputs in both conditions. The two incomplete requests terminate in the failure-label writer before assistant text is produced and are excluded from the paired effect calculation. Among the four complete pairs, every pair changes normalized memory content and the extracted memory-item title set. Token-set Jaccard distance is 0.691 for task 21 and 0.708 for task 22. It is 0.770 for both task 23 and task 25, giving mean distance 0.735. The 4/4 change rate is therefore an existence result conditional on paired completion. Panel (a) of Figure 2 shows the complete pair-level result.

The content changes are procedural as well as lexical. For task 21, success-conditioned memory centers on entering the review section and preserving direct quotations, whereas failure-conditioned memory emphasizes exhaustive pagination and extraction verification. For task 23, the success writer emphasizes evidence collection across pages while the failure writer emphasizes pagination state and broader synonym coverage. To make this distinction explicit without introducing another learned judge, we reuse a deterministic operation-slot taxonomy over the frozen memory-item titles: navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, verification/completeness, temporal filtering, and status filtering. The slot set changes in all four complete pairs; mean slot-set Jaccard distance is 0.493 (range 0.400–0.571). Moreover, every

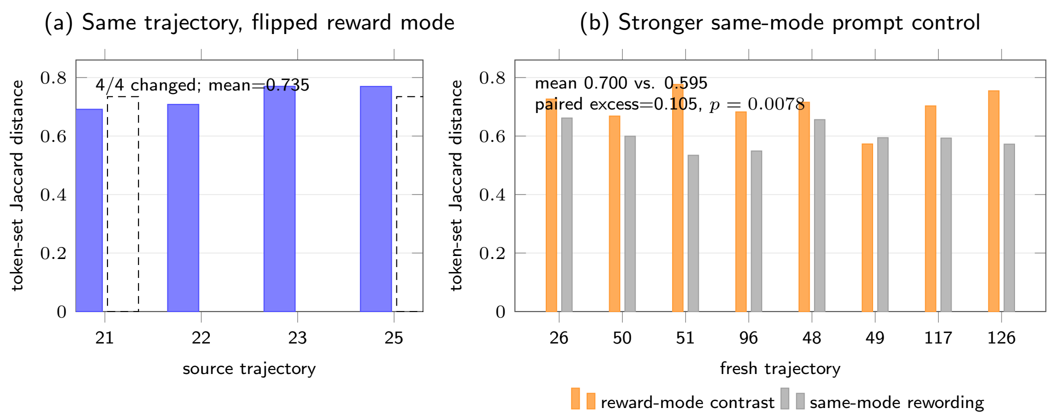

Figure 2: Write-channel evidence and its strongest simple control. (a) Among the four paired write interventions complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

complete pair has a success-only slot in evidence capture, navigation, or query expansion, while verification/completeness appears as a failure-only slot in three of four pairs. This is a structural diagnostic rather than embedding-level semantic proof; Section 6 asks whether the same structure also changes under reward-preserving prompt rewording.

This paired result establishes the first link of the mechanism: changing the reward-conditioned writer branch changes persistent memory under a fixed experience. Because the branch itself changes the writer instruction, the next section tests the strongest simple alternative explanation: generic sensitivity to prompt wording.

6 Ruling Out Generic Prompt Sensitivity

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from the paired write sample and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8}\sum_{i=1}^{8}(B_i - W_i). \tag{4}$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

The same conclusion is visible at the procedural-structure level. Without changing the operation-slot taxonomy used in the paired write intervention, we apply it to the frozen title sets from all eight control trajectories. Mean slot-set distance between the released success and failure modes is 0.556, compared with 0.247 for semantic-preserving rewordings within the same mode, an average structural excess of 0.309. Five tasks have positive excess, two tie, and one is negative. We treat this as a descriptive controlled diagnostic, not a second significance test or an embedding-semantic claim. Its role is narrower: the structural shift associated with reward mode is larger on average than the structural shift produced by reward-preserving wording changes under the same fixed taxonomy.

7 Does the Persistent-State Perturbation Reach Future Behavior?

The write-channel result matters only if the persistent state difference survives reuse. We test this in two stages. The first asks whether paired memories can change an immediate decision under an identical browser observation. The second measures terminal success under a fully crossed source-memory by future-task design.

7.1 Fixed-state action reach

The four complete write-stage pairs are mapped to four held-out Shopping states. Within each state, the future task prompt and browser observation are identical across memory conditions. The policy model and output schema are fixed. The original action-reach experiment contains 12 aligned success-memory versus failure-memory rollouts; 7 pairs choose different structured next actions, giving divergence rate 0.583 and passing the frozen existence falsifier. This establishes an action-level witness: paired memories can change the next decision under the same observation.

We separately stress-test the stronger claim that the first-action distributions are systematically separated across the frozen state set. This test uses eight rollouts per condition on every frozen state. Mean empirical total-variation distance between first-action distributions is 0.344. The 100,000-draw stratified permutation audit gives $p = 0.311$, so this stronger distribution-level test is a negative result: the available state set does not resolve systematic first-action distribution separation. The action-level evidence therefore rests on the paired existence witness (7/12 divergent next actions), not on a population-level distribution claim. Panel (a) of Figure 3 places this negative stress test beside the positive terminal evidence.

7.2 Fully crossed terminal confirmation

Terminal evaluation uses all four complete source-memory pairs and four future tasks frozen before the first terminal-policy call. Every source pair is crossed with every future task, producing 16 cells. Within a cell, released browser evidence is byte-identical across memory conditions. Doubao Seed 2.0 Mini generates the terminal answer at temperature 0.2, and the original deterministic WebArena string-match evaluator scores success.

The initial frozen run uses three rollouts per memory condition. All 96 primary calls complete, with mean absolute success-rate difference 0.145833 and permutation $p = 0.160128$, so its dual gate does not pass. This non-pass was already committed as `730c2bc` before the confirmatory contract and before any confirmatory provider call. The confirmatory stage is therefore a targeted *uniform replication after the initial non-pass*, not the first outcome-blind terminal experiment: it keeps the same four source memories, same four future tasks, same 0.15 effect floor, and same $p < 0.05$ gate, forbids source/task selection after outcomes, and increases only the fresh replication depth from three to eight per condition in every cell. All 256 confirmatory calls complete. Mean absolute success-rate difference is 0.15625 and the 100,000-draw within-cell permutation test gives $p = 0.00074$. Appendix 3 reports all 16 initial and confirmatory cells side by side. Eight confirmatory cells have zero observed contrast; six signed effects are positive and two negative. The two largest cells, 22/388 and 25/387, have signed effects 0.750 and 0.625 and account for 78.2% of squared-effect mass. We therefore interpret the result as finite distributional sensitivity, not a stable per-task directional effect. Because the two provider-incomplete source pairs are absent from this matrix, the result is conditional on the four completed source pairs. Panel (b) of Figure 3 shows the confirmatory signed matrix.

7.3 Fresh no-memory branch-location control

To locate the two confirmed memory branches relative to omission without changing the confirmatory gate, we freeze a source-independent no-memory control with eight fresh rollouts on each of the same four future tasks. All 32 recovery calls complete. No-memory success rates are 1.0, 0.0, 0.0, and 1.0 for future tasks 164, 385, 387, and 388. The same future-task baseline is shared across four source comparisons and is therefore not counted four times as independent evidence; we report Wilson intervals and cellwise Newcombe contrasts but no new global p-value. At the point estimate, omission matches both memory branches in eight cells, is closer to the success-memory branch in six, and closer to the failure-memory branch in two. For source/future 22/388, success-memory, failure-memory, and no-memory rates are 0.25, 1.0, and 1.0: the success-conditioned state is the branch that departs from omission. For 25/387, the corresponding rates are 0.125, 0.75, and 0.0, so both memories improve over omission but by different amounts. Thus the confirmed reward-conditioned contrast cannot be reduced to "failure memory is harmful" or "success memory is helpful"; which branch moves, and in which direction, is source–future specific.

7.4 What transfers across writers?

A sequential GLM-5.3 replication holds source trajectories, ReasoningBank reward-mode prompts, future-task support, downstream policy, evaluator, and terminal gate fixed. After one preregistered output-cap repair, all eight writer calls complete; all four pairs change content and title sets, with mean token-set Jaccard distance 0.737. The unchanged 4-by-4 terminal stage then completes 256/256 scorable units. Mean absolute success-rate difference is 0.140625 with $p = 0.00012$: the permutation criterion passes, but the effect misses the frozen 0.15 floor by 0.009375, so cross-writer terminal generalization is not established. Among six cells nonzero under both writers, four retain sign and two reverse (22/388: $+0.75 \rightarrow -0.25$; 23/387: $+0.125 \rightarrow -0.75$). Thus the write-time channel replicates more robustly than downstream effect magnitude or direction.

Where does the heterogeneity sit? We add a descriptive two-way decomposition of the centered signed 4-by-4 effect matrix, using only the already frozen confirmatory cells. Source-memory main effects account for 9.4% of centered sum-of-squares and future-task main effects for 6.5%; the remaining 84.1% is source–future interaction residual. The interaction is visible without modeling: source 21 is positive on future tasks 385/387 but negative on 388, while source 22 is positive on 385/388 but negative on 387. Conversely, future tasks 387 and 388 change sign across source memories. Thus the large effects are not well summarized as a fixed "bad memory" or a fixed "hard task" effect in this matrix. Outcome headroom also constrains what can be observed: future task 164 is at a joint 1.0/1.0 ceiling for every source pair and contributes zero squared-effect mass, whereas all observed squared-effect mass lies in the three non-ceiling future tasks. Write-stage divergence magnitude does not give a simpler explanation. Sources 23 and 25 have nearly identical token-set distances (0.770 in both cases) and identical operation-slot distance (0.500), yet their mean absolute terminal effects across the four future tasks are 0.031 and 0.156, respectively; source 22 has a smaller token distance (0.708) but the largest source-averaged effect (0.250). Thus "more different memory" is not a monotonic proxy for stronger downstream sensitivity in these four sources. Because task identity, workflow, binary-outcome headroom, and source–future compatibility are confounded in this finite matrix, these diagnostics characterize the observed heterogeneity rather than define a predictive transfer model.

The signed failure-memory minus success-memory effect averages 0.09375, while individual cells have both signs. Variance amplification therefore does not require every reward error to be uniformly harmful. It requires reward-conditioned memory state to alter conditional outcome distributions across runs.

7.5 Bounded consequence analysis under hypothetical reward corruption

Let $p_S(c)$ and $p_F(c)$ denote the confirmed success probabilities for success-conditioned and failure-conditioned memory in frozen cell c. If reward-label corruption selects the failure-conditioned memory with probability q, the between-memory success variance in that cell is

$$V_c(q) = q(1-q)\left[p_F(c) - p_S(c)\right]^2 . \tag{5}$$

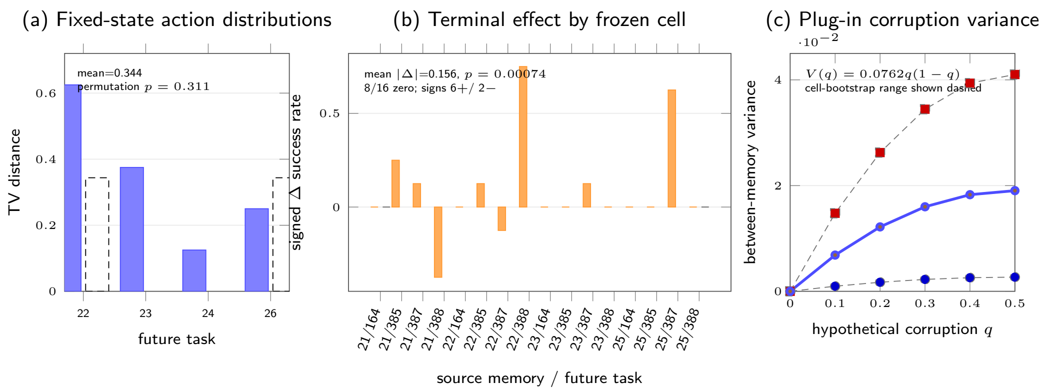

Figure 3: Downstream evidence for persistent-state propagation. (a) Across four frozen states with eight rollouts per memory condition, the larger first-action distributional stress test does not pass its permutation gate ($p = 0.311$). (b) The 16-cell matrix crosses all four complete source-memory pairs with four frozen future tasks; signed terminal effects reveal substantial heterogeneity, with aggregate mean absolute difference 0.15625 and $p = 0.00074$. (c) A plug-in decomposition gives the between-memory variance curve for hypothetical reward-label corruption probability q; dashed curves show the 95% cell-resampling sensitivity range.

This curve is a deterministic plug-in decomposition of the measured conditional effects, not an additional corruption-sweep rollout experiment. Across all 16 confirmatory cells, the mean squared conditional effect is 0.0762. Averaging over cells gives

$$\bar{V}(q) = 0.0762\, q(1 - q). \tag{6}$$

A 100,000-repetition nonparametric bootstrap that resamples the 16 frozen cells gives a descriptive 95% sensitivity interval $[0.0107, 0.1641]$ for the mean squared effect. At $q = 0.5$, the plug-in estimate is 0.0190 with corresponding cell-resampling interval $[0.0027, 0.0410]$. The preregistered within-cell permutation test remains the primary confirmatory significance test. Figure 3(c) shows the plug-in curve together with this finite-cell sensitivity range.

8 Limitations

The write-stage estimate is conditional on paired provider completion. Six source trajectories are requested, but two failure-branch responses terminate before assistant text exists; GET-only recovery returns no memory text. Receipt-level forensics distinguish these missing arms from downstream parsing failure, but cannot rule out a label-dependent provider completion process or identify the direction of any resulting selection. We therefore report 4/4 divergence only among complete pairs and carry only those four source-memory pairs into downstream analysis.

The downstream evidence is intentionally finite and fixed-evidence. The larger first-action distribution stress test is unresolved ($p = 0.311$), so the action claim remains an existence witness rather than population-level separation. The terminal result comes from a same-support uniform replication after an initial non-pass, retaining the identical four sources, four future tasks, 0.15 practical-effect floor, and $p < 0.05$ gate while increasing only replication depth. Although the 256-call replication passes that gate, 8/16 cells are zero and two cells contain 78.2% of squared-effect mass. The result therefore supports heterogeneous conditional sensitivity on this frozen matrix, not a stable task-level direction or a population effect. Source-faithful live browser navigation remains unexecuted because the released Shopping endpoint and reset service are unreachable from the execution hosts; this is transport debt, not negative scientific evidence.

Cross-writer evidence separates the write mechanism from downstream transport. GLM-5.3 changes all four paired memories and title sets, but its terminal replication yields 0.140625, below the preregistered 0.15 floor despite $p = 0.00012$, with two direction reversals among six cells nonzero under both writers. Thus the write-time state intervention has bounded support across two writer families, while writer-invariant downstream magnitude or direction is not established. Additional writers and task domains are needed before making broader claims.

The no-memory arm is a source-independent branch-location control, not an independent $4 \times 4 \times 3$ factorial: each future-task baseline is reused descriptively across source rows, so we add no global significance test. Likewise, the corruption-rate curve is a deterministic consequence analysis of measured conditional effects, not an empirical corruption sweep. Under the released ReasoningBank retriever, label-invariant task descriptions are embedded with top-1 retrieval at threshold 0.3 and only the selected entry is injected; nonselected corruption-mask bits are therefore inert. A genuine multi-memory corruption interaction would require a different retrieval substrate. Token-set Jaccard and the fixed operation-slot taxonomy remain coarse diagnostics rather than learned semantic measures.

9 Conclusion

We ask a narrow causal question: when terminal feedback chooses how an episode is written into external memory, can an erroneous label become durable agent state even though the underlying experience is unchanged? On byte-identical released trajectories, switching only the reward-conditioned reflection branch changes every complete paired memory. A stronger same-mode semantic-preserving wording perturbation does not absorb the effect, supporting a reward-conditioned write channel rather than generic prompt sensitivity alone.

That state difference can reach later behavior under matched future evidence. Paired memories change 7/12 aligned next actions, and a same-support 4×4 terminal replication after an initial non-pass yields mean absolute success-rate difference 0.15625 with permutation $p = 0.00074$. The effect is heterogeneous, and a no-memory control rules out a uniform "success good / failure bad" interpretation. GLM-5.3 reproduces the upstream write divergence but misses the frozen downstream practical-effect floor, so the evidence is stronger for the existence of the write-time channel than for invariant downstream magnitude or direction.

The resulting implication is configuration-bound but operationally important. In the released ReasoningBank/WebArena Shopping mechanism, terminal evaluation is part of state update: its label selects what durable memory is written, and that state can alter later outcome distributions. For reward-conditioned external-memory agents of this form, evaluator reliability is therefore also a persistent-state reliability requirement. Broader claims about writers, domains, live interaction, or corruption processes require separate evidence rather than extrapolation from this finite controlled study.

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A F0 Reproduction Details

A.1 Released trajectory sample

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 Memory construction intervention

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

A.3 Pair-level F0 results

Table 1: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B Prompt-Perturbation Control Details

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 2 reports the paired distances used in the frozen sign-flip test.

Table 2: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C Downstream Confirmatory Details

C.1 Fixed-state action distributions

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 Frozen terminal matrix

The initial terminal experiment and F2R1 use the same complete F0 source set $\{21, 22, 23, 25\}$, the same future-task set $\{164, 385, 387, 388\}$, and the same aggregate absolute-effect statistic with the same dual gate: mean absolute difference at least 0.15 and within-cell permutation $p < 0.05$. The initial stage uses three fresh rollouts per condition per cell and is already frozen in Git commit `730c2bc` at 2026-08-22 00:01:42 +08:00; it yields 0.145833 with $p = 0.160128$ and does not pass. The separately authorized F2R1 contract is content-addressed as `ea9a2260...01f5`; its first provider receipt is generated at 2026-08-22 01:50:32 +08:00. It explicitly forbids source/future-task selection after outcomes and increases only uniform replication depth to eight fresh rollouts per condition in every cell. All 256 F2R1 calls complete, yielding 0.15625 with $p = 0.00074$. Thus F2R1 is a targeted same-support replication after an initial non-pass, not a claim that the confirmatory design was chosen before any terminal outcomes were observed.

Table 3: Initial F2 and F2R1 cell-level terminal rates on identical 4-by-4 support. S/F denote success-/failure-label memory.

Source	Future	F2 S	F2 F	F2 $\|\Delta\|$	F2R1 S	F2R1 F	F2R1 $\|\Delta\|$
21	164	1.000	1.000	0.000	1.000	1.000	0.000
21	385	0.000	0.333	0.333	0.000	0.250	0.250
21	387	0.000	0.333	0.333	0.125	0.250	0.125
21	388	1.000	0.333	0.667	1.000	0.625	0.375
22	164	1.000	1.000	0.000	1.000	1.000	0.000
22	385	0.000	0.667	0.667	0.000	0.125	0.125
22	387	0.000	0.000	0.000	0.125	0.000	0.125
22	388	1.000	1.000	0.000	0.250	1.000	0.750
23	164	1.000	1.000	0.000	1.000	1.000	0.000
23	385	0.000	0.000	0.000	0.000	0.000	0.000
23	387	0.000	0.000	0.000	0.125	0.250	0.125
23	388	0.667	1.000	0.333	1.000	1.000	0.000
25	164	1.000	1.000	0.000	1.000	1.000	0.000
25	385	0.000	0.000	0.000	0.000	0.000	0.000
25	387	0.000	0.000	0.000	0.125	0.750	0.625
25	388	1.000	1.000	0.000	1.000	1.000	0.000
Mean $\|\Delta\|$		0.145833			0.15625		

C.3 Fresh no-memory branch-location control

The no-memory control is frozen after F2R1 as a secondary branch-location diagnostic. It uses the identical four future tasks, released evidence packets, Doubao Seed 2.0 Mini settings, evaluator, and eight-rollout depth, but no source-memory dimension. The earlier 12 exploratory no-memory calls are excluded. An initial 32-call execution is also excluded from science because a new local validator incorrectly required the provider's resolved model name to equal the requested alias; all 32 responses resolved to `doubao-seed-2-0-mini-260215`, which is exactly the versioned model name recorded in all 256 historical F2R1 receipts. After freezing that execution-layer diagnosis and archiving provider responses before local validation, the one-for-one recovery completes 32/32 calls. Both attempts are counted in execution cost; only the recovery outcomes enter the analysis.

Table 4: Fresh no-memory terminal baseline. Wilson intervals are per future task and are not replicated across source memories.

Future task	Successes	Rate	Wilson 95% interval
164	8/8	1.000	[0.676, 1.000]
385	0/8	0.000	[0.000, 0.324]
387	0/8	0.000	[0.000, 0.324]
388	8/8	1.000	[0.676, 1.000]

Combining these four source-independent baselines descriptively with the 16 F2R1 cells yields eight cells where omission matches both memory branches, six where omission is closer to the success-memory branch, and two where it is closer to the failure-memory branch. The largest illustrative contrasts are source/future 22/388, where $(p_S, p_F, p_0) = (0.25, 1.0, 1.0)$, and 25/387, where $(p_S, p_F, p_0) = (0.125, 0.75, 0.0)$. The first isolates the success-conditioned memory as the deviation from omission; the second places both memories above omission. Because each p_0 is shared by four source rows, we report cellwise score intervals and no new global significance test.

C.4 Cross-writer replication and frozen negative boundary

The writer-family test is sequential. Stage 1 first rewrites only source tasks 21, 22, 23, and 25 under the same released success/failure ReasoningBank prompts using GLM-5.3 at temperature zero. The parent 2,200-token attempt is scientifically invalid after two failure-mode responses terminate at the exact output cap with no assistant text; an overlapping-runner race also makes its provider POST count reconstructible only as a lower bound of nine. The single preregistered repair changes only the output cap to 4,096 and adds a single-writer transaction lock. Its eight calls all complete; all four pairs change content and title sets, with token-set distances 0.783, 0.728, 0.715, and 0.723 (mean 0.737482).

Stage 2 keeps the original Doubao Seed 2.0 Mini downstream policy, four future tasks, evidence packets, evaluator, eight-rollout cell depth, and dual terminal gate unchanged. All 256 cells-by-condition rollouts are scorable. The observed mean absolute effect is 0.140625 and the 100,000-draw permutation test gives $p = 0.00012$. The statistical gate passes but the frozen 0.15 minimum-effect gate does not, so the registered cross-writer terminal generalization claim is not established. Table 5 shows that the difference is not a uniform attenuation.

Among the six cells that are nonzero under both writers, four retain direction and two reverse. Reusing the already frozen writer-independent no-memory rates only as descriptive baselines yields 10 GLM cells equidistant from omission, four closer to the success-memory branch, and two closer to the failure-memory branch; no additional provider calls or global test are introduced. This supports the upstream write-channel replication while preserving the failed terminal-invariance gate.

C.5 Heterogeneity and corruption-rate decomposition

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the paper treats the result as a variance channel instead of a uniform directional harm effect.

If reward-label corruption selects the failure-conditioned memory with probability q, the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell c. The exact mean squared conditional effect

Table 5: Signed failure-minus-success terminal effects under the original DeepSeek-V4-Flash writer (D) and the GLM-5.3 writer (G).

Source	Future	D effect	G effect
21	164	0.000	0.000
21	385	0.250	0.250
21	387	0.125	0.375
21	388	-0.375	0.000
22	164	0.000	0.000
22	385	0.125	0.000
22	387	-0.125	-0.125
22	388	0.750	-0.250
23	164	0.000	0.000
23	385	0.000	0.000
23	387	0.125	-0.750
23	388	0.000	0.000
25	164	0.000	0.000
25	385	0.000	0.000
25	387	0.625	0.500
25	388	0.000	0.000

across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

C.6 Controlled structural diagnostic

The controlled structural diagnostic reuses the exact operation-slot taxonomy already frozen for the F0 title audit; it does not add a learned semantic judge. Applied to the eight disjoint prompt-control trajectories, the mean slot-set distance between released reward modes is 0.555655, whereas the mean distance induced by semantic-preserving rewording within the same mode is 0.246578. The mean between-minus-within structural excess is 0.309077; five tasks are positive, two tie, and one is negative. This is descriptive supporting evidence rather than a second hypothesis test. On the four complete F0 pairs, success-only operation slots include at least one of evidence capture, navigation, or query expansion in all four pairs, while verification/completeness is failure-only in three pairs. Provider receipts also show longer failure-mode responses in all four complete pairs (mean output-token difference +266.5; mean reasoning-token difference +268.0), which we report only as response-structure asymmetry, not as semantic quality.

C.7 Source–future interaction diagnostic

A two-way additive decomposition of the centered signed F2R1 cell effects is shown below. It is a finite-matrix description, not inferential ANOVA.

Table 6: Descriptive decomposition of centered signed terminal effects.

Component	Sum of squares	Share
Source-memory main effect	0.101562	9.4%
Future-task main effect	0.070312	6.5%
Source $\times$ future interaction residual	0.906250	84.1%
Total	1.078125	100%

The source and future marginals therefore summarize little of the signed effect structure. Source 21 is positive on future tasks 385 and 387 but negative on 388; source 22 is positive on 385 and 388 but negative on 387. Future tasks 387 and 388 likewise change sign across source memories. Table 7 adds benchmark-observable future-task attributes. Future task 164 is at a joint 1.0/1.0 terminal ceiling

Table 7: Future-task attributes and concentration of the frozen terminal effect. "Home" means the task starts at the Shopping root rather than a product page.

Task	Start	Ref. items	Mean $\lvert\Delta\rvert$	Squared-mass share
164	Product	2	0.00000	0.0%
385	Home	8	0.09375	6.4%
387	Home	6	0.25000	35.9%
388	Home	2	0.28125	57.7%

for every source pair, so it has no observable binary-outcome headroom and contributes zero effect mass. The other three tasks contain all observed squared-effect mass. With only four future tasks, start context, task identity, and outcome headroom are confounded; these rows characterize where the frozen effect occurred but do not define a general transfer predictor.

A second descriptive check asks whether sources whose paired memories differ more at write time simply create larger downstream effects. Table 8 shows that this reduction is not supported by the four completed sources. Sources 23 and 25 are effectively matched on both reported write-divergence

Table 8: Write-stage divergence versus source-averaged terminal sensitivity. The final column averages $|\Delta|$ over the four frozen future tasks.

Source	Token distance	Slot distance	Mean terminal $\lvert\Delta\rvert$
21	0.691	0.571	0.188
22	0.708	0.400	0.250
23	0.770	0.500	0.031
25	0.770	0.500	0.156

magnitudes but differ fivefold in source-averaged terminal sensitivity. Across the four points, the descriptive Pearson correlations are -0.730 for token distance and -0.367 for slot distance versus mean terminal $|\Delta|$. With $n = 4$ these correlations have no inferential status; they are included only to reject the simple monotonic reading that a larger textual or operation-slot perturbation necessarily produces a larger downstream effect.

D Execution Accounting and Reliability

All scientific evidence in this paper is inference-only; there are no training runs or local GPU fine-tuning runs. Directly countable execution comprises 12 primary write requests (10 complete), 32 prompt-control writer requests, 96 first-action distribution calls, 108 initial terminal calls, 256 confirmatory terminal calls, 64 no-memory calls (32 execution-invalid plus 32 scientifically usable recovery calls), eight repaired GLM writer calls, and 256 GLM terminal calls. These stages account for 832 provider requests outside the failed parent GLM writer attempt. That parent attempt has zero scientific authority and an exact POST count that is not reconstructible after an overlapping-runner race; its observable lower bound is nine. The full provider-POST lower bound is therefore at least 841, excluding the unresolved low-level request count of the earlier 12-unit aligned action-existence witness. Provider responses are archived before analysis where the frozen runner supports it; incomplete or execution-invalid units are never imputed into scientific estimates.